There are multiple ways that one can prove that 2 triangles are congruent.

Acronyms
Let's get familiar with the acronyms used.

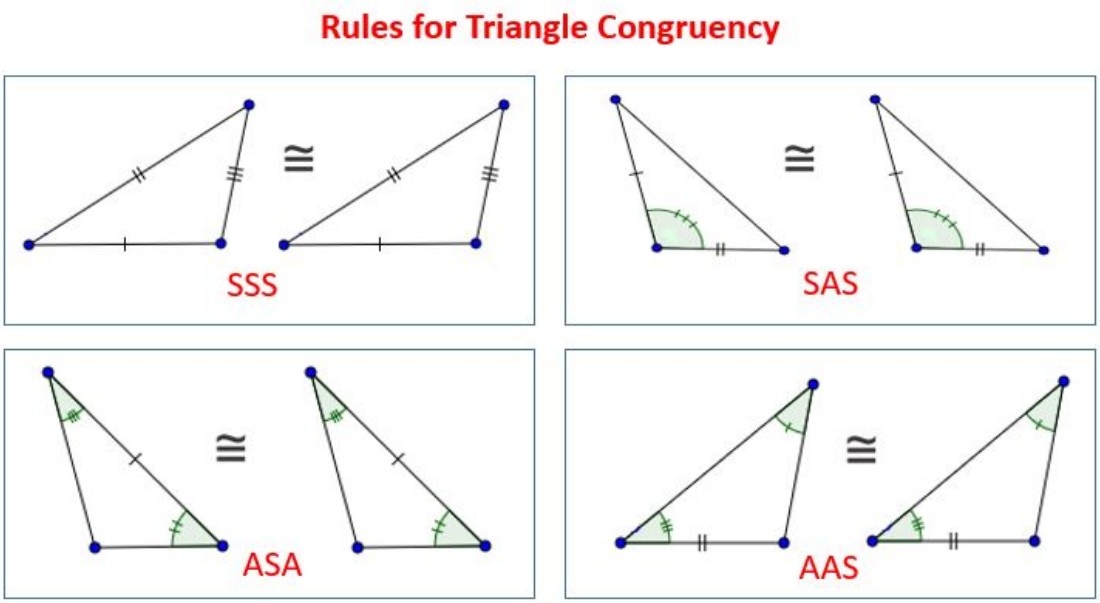
"A" means angle
This would mean that you know that an angle on one triangle is the same as another angle on another triangle.

"S" means side
This would mean that a side on one triangle is the same length as a side on another triangle.

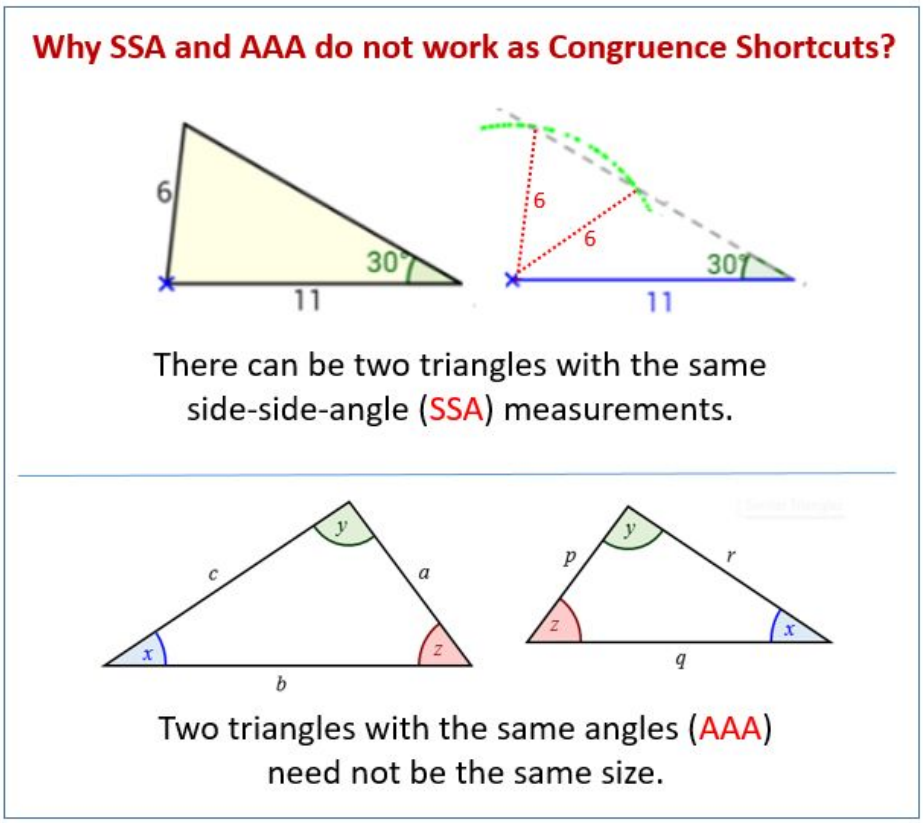
Not an acronym but a symbol, $\cong$, means congruent.

Combinations of these get us congruence problems. But, not all combinations work. Here is a list of the ones that work and don't work:

Valid	Not valid
• SSS • SAS • ASA • AAS	• AAA • SSA (please be mature)

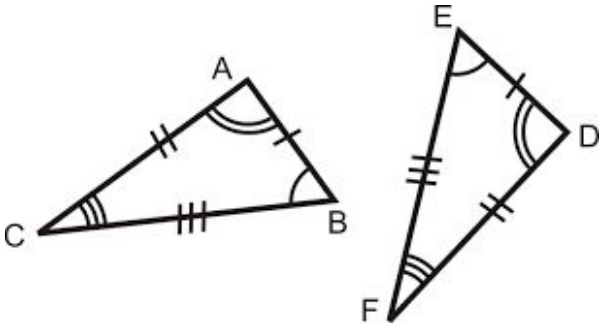


(**note**: the corresponding hatches made on the triangles signify which sides/angles are the same. In the first picture, the sides with 1 line through them mean that those 2 sides are the same. The same goes for angles, in the picture titled "ASA", the angles with 2 lines on them signify that those 2 angles are of the same value.)



Writing a Congruence Statement

To make sure we communicate that one triangle is congruent to another triangle, we have to write it's label in a certain order.

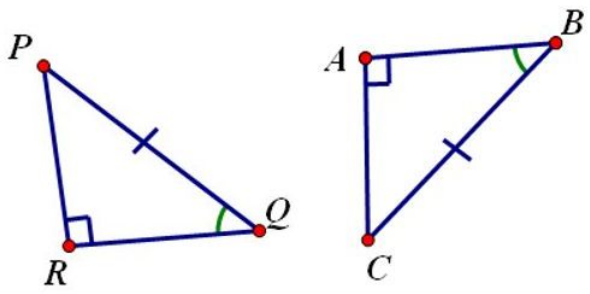


https://www.ck12.org/geometry/triangle-congruence/lesson/Congruent-Triangles-BSC-GEOM/

These triangles are congruent according to SSS, SAS, ASA, and AAS. Now to label it correctly we need to correspond each congruent side and angle.

$$\overline{AB} \cong \overline{DE} \qquad \angle A \cong \angle D$$
$$\overline{BC} \cong \overline{EF} \text{ and } \angle B \cong \angle E$$
$$\overline{AC} \cong \overline{DF} \qquad \angle C \cong \angle F$$

So $\Delta ABC \cong \Delta DEF$



https://www.onlinemath4all.com/images/xtrianglecongruenceandsimilarity3.png.pagespeed.ic.JFiTK0NPPL.png

These 2 triangles can be proved congruent with AAS (angle-angle-side).
Since, $\angle R \cong \angle A$, $\angle Q \cong \angle B$, and $\overline{PQ} \cong \overline{BC}$
Then $\Delta PQR \cong \Delta CBA$

(**Note:** When we wrote out the congruent statement, each angle of the triangle matched with its congruent one on the other triangle)

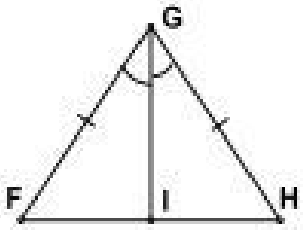
$\Delta PQR \cong \Delta CBA$

Now we know more about triangle congruence, let's practice!

Practice problems

For each pair to triangles, state the postulate or theorem that can be used to conclude that the triangles are congruent.

1. _____



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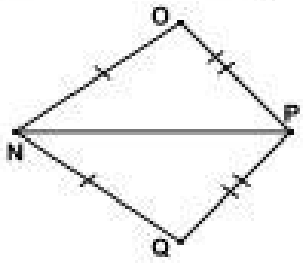

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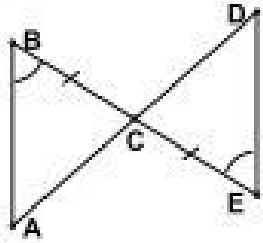

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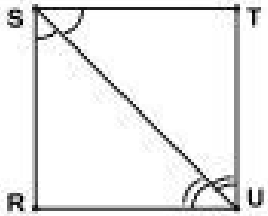

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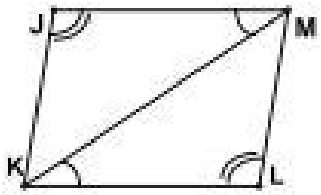

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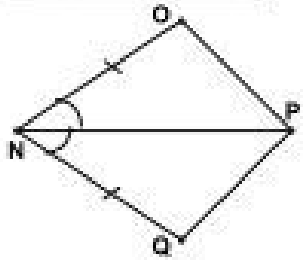

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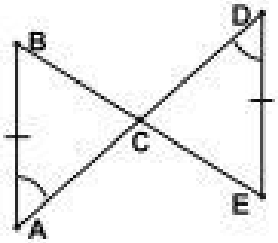

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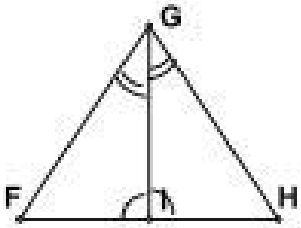

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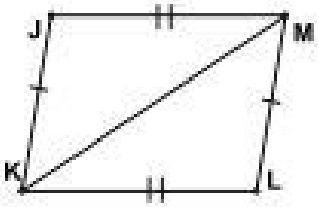

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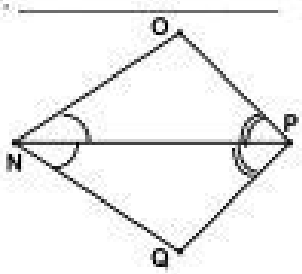

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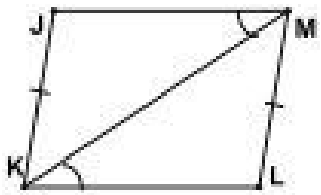
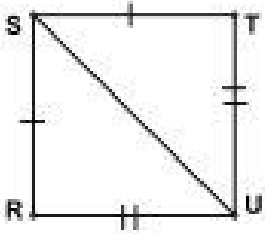
12. _____